

Diabetic Nephropathy

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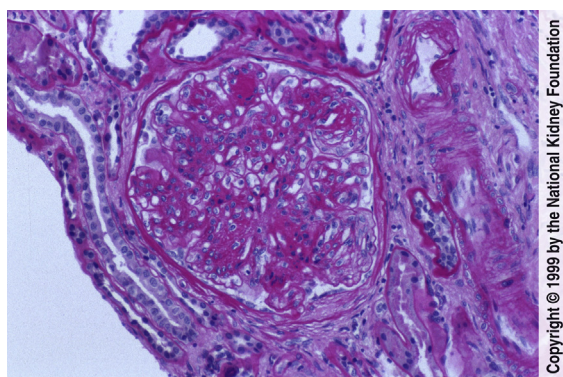


Fig 1. Diabetic nephropathy is characterized by light microscopy by mesangial expansion, sometimes with nodule formation, mesangial hypercellularity, and a thickened glomerular basement membrane. In this case, there is diffuse mesangial hypercellularity and expansion with only small nodule formation. (Periodic acid-Schiff stain, original magnification $\times 200$).

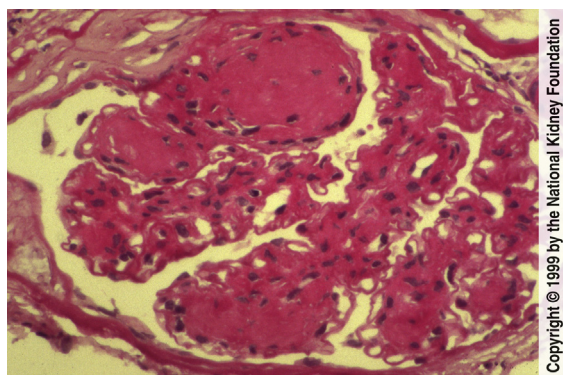


Fig 2. Diabetic nephropathy is one of the causes of nodular glomerulosclerosis, illustrated here, with large nodules of matrix within mesangial areas with lesser increase in mesangial cellularity. The glomerular basement membrane is thick without apparent deposits. (Periodic acid-Schiff stain, original magnification $\times 400$).

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Am J Kidney Dis. 34(5):E18-E19.1.

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0272-6386/\$36.00

[http://dx.doi.org/10.1053/S0272-6386\(13\)90022-6](http://dx.doi.org/10.1053/S0272-6386(13)90022-6)

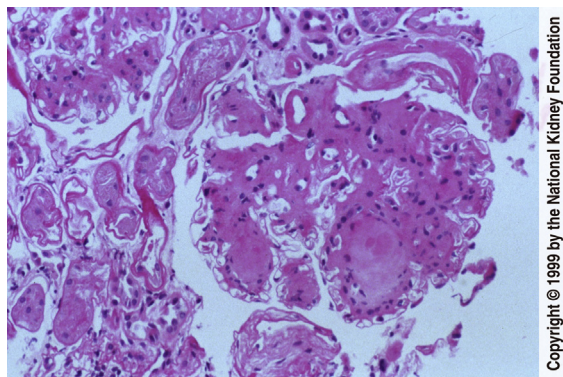


Fig 3. Nodular glomerulosclerosis due to diabetic nephropathy may present with large nodules of relatively acellular mesangial matrix, as illustrated here. The glomerular basement membrane is thickened, and both afferent and efferent arterioles are hyalinized, in contrast to the hyalinization of the afferent arteriole only in hypertension. (Periodic acid-Schiff stain, original magnification $\times 200$).

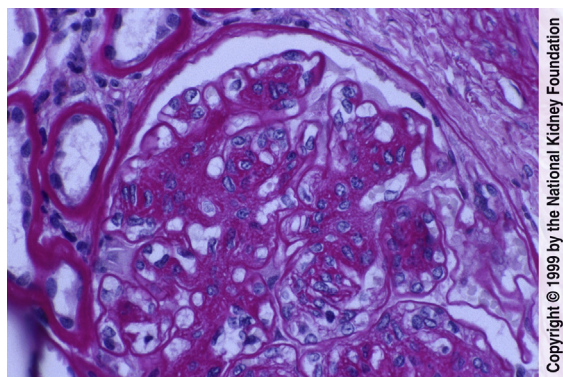


Fig 4. There is diffuse mesangial matrix expansion and increased mesangial hypercellularity and prominent glomerular basement membranes in diabetic nephropathy, as shown here. The basement membrane is uniformly thick without evident deposits. (Periodic acid-Schiff stain, original magnification $\times 400$).

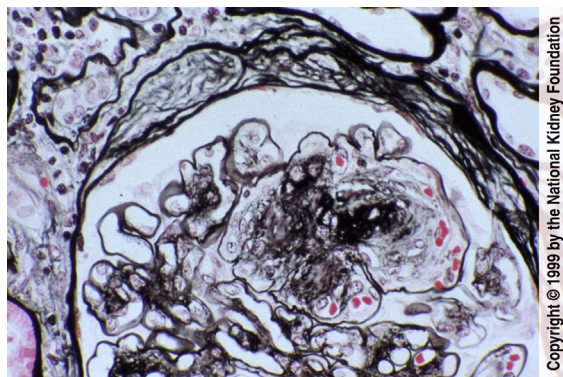


Fig 5. The glomerular basement membrane is thickened without evident deposits in diabetic nephropathy. The nodular mesangial expansion has a lamellated appearance on this silver stain, which is thought to be due to repeated injury with mesangiolysis and subsequent increased matrix accumulation. RBC fragments are present in the nodule, likely a consequence of local microvascular injury. (Jones' silver stain, original magnification $\times 400$).

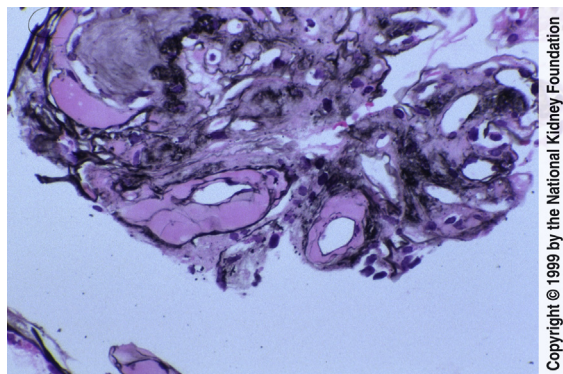
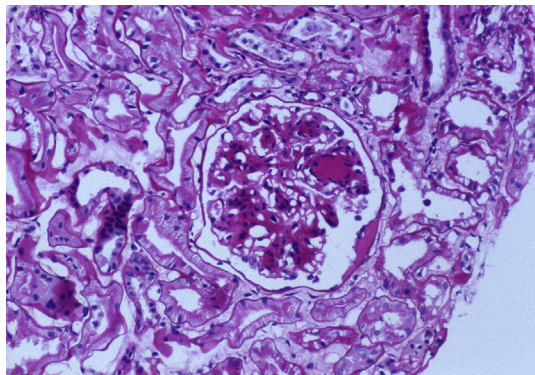
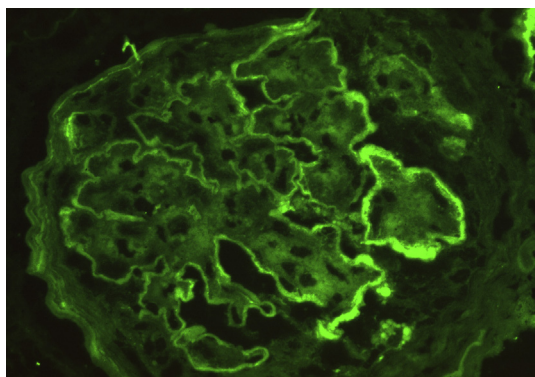


Fig 6. Both afferent and efferent arterioles, shown here at the vascular pole, are hyalinized in diabetic nephropathy. This insudation of plasma proteins is due to endothelial injury. In contrast to hypertensive lesions where only the afferent arteriole is affected, there is injury to both afferent and efferent arterioles in diabetic nephropathy. There is also mesangial matrix expansion and prominent basement membranes, and a focus of hyaline within the sclerotic area in the glomerulus (top left). (Jones' silver stain, original magnification $\times 400$).



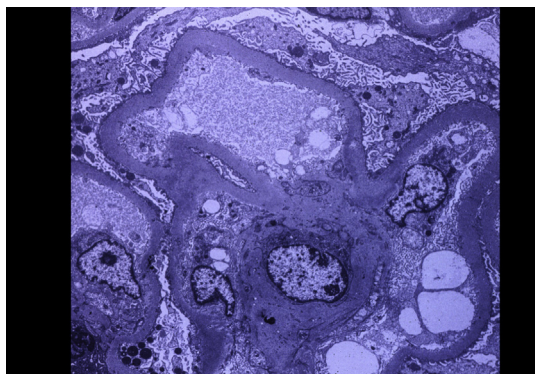
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Fig 7. Diabetic nephropathy may also show hyalin within Bowman's capsule, the so-called capsular drop. This lesion is uncommon and, although not entirely specific for diabetic nephropathy, is distinctly unusual in other entities. The glomerulus shows mesangial expansion, a small area of nodule formation, and prominent basement membranes, all indicative of diabetic nephropathy. (Periodic acid-Schiff stain, original magnification $\times 100$).



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Fig 8. This case shows nodular glomerulosclerosis and linear accentuation of the glomerular basement membrane with IgG, which is typical of diabetic nephropathy. There are no immune complexes in diabetic nephropathy, although obviously superimposed disease may occur (IgA nephropathy, membranous glomerulonephritis, and thin basement membrane lesion are particularly common as concurrent glomerular lesions). It is also important to rule out light chain deposition disease in cases of nodular glomerulosclerosis, which may be done by examining k and l staining. (Anti-IgG immunofluorescence, original magnification $\times 400$).



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Fig 9. Electron microscopic studies in diabetic nephropathy show increased mesangial matrix and cellularity without deposits, and a thickened glomerular basement membrane due to marked widening of the lamina densa without deposits. Foot processes frequently show effacement, most often subtotal. Tubular basement membranes do not show deposits, an important feature in excluding possible light chain deposition disease in cases of nodular glomerulosclerosis. Occasional fibrillar collagen may be present in sclerotic mesangial areas in diabetes. (Transmission electron microscopy, original magnification $\times 3,000$).